

scRNA-seq Workshop Part 1

Applied Bioinformatics for Genomics

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scRNAseq versus Bulk RNA-seq



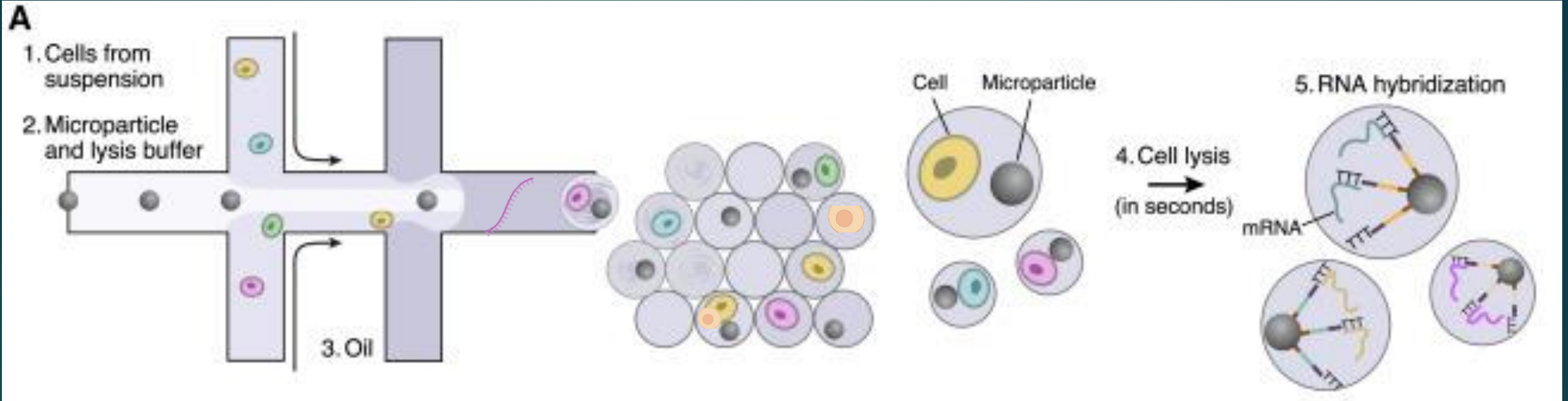
Bulk RNA-seq
averages across the
population



scRNA-seq reports
per-cell expression
and enables
computational
“sorting”

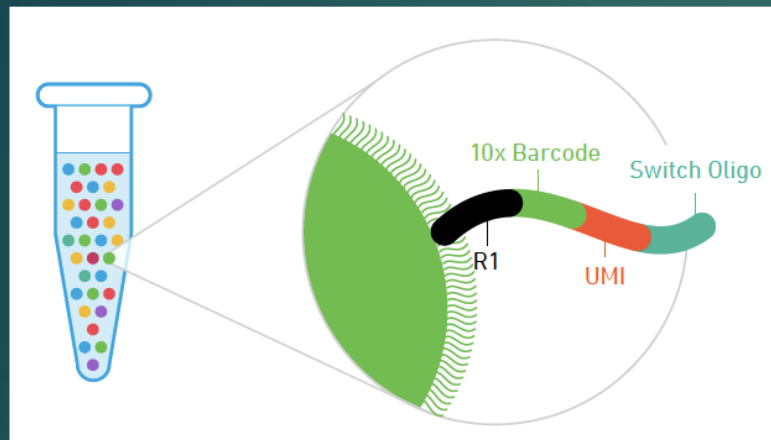


Droplet based scRNA-seq: The Good & The Ugly

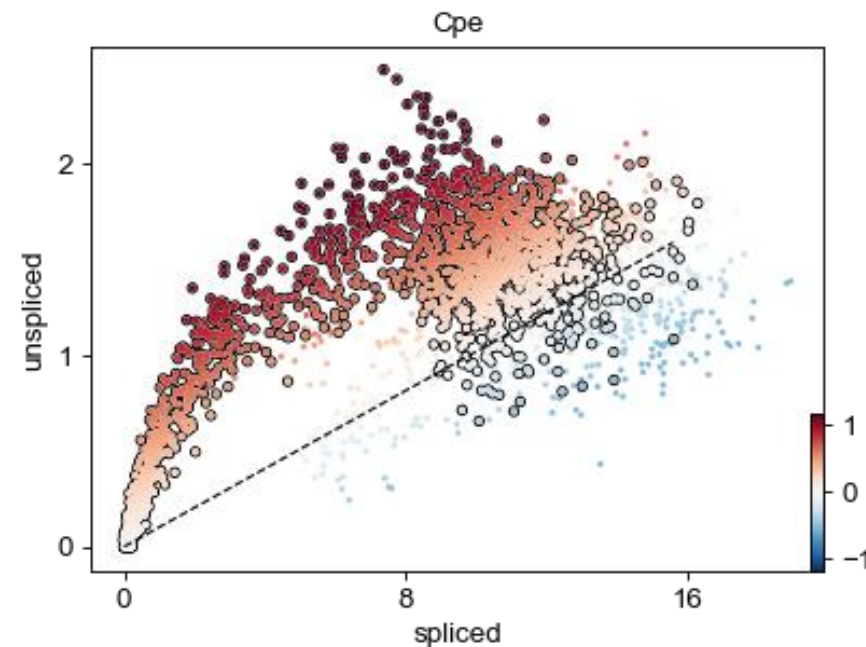
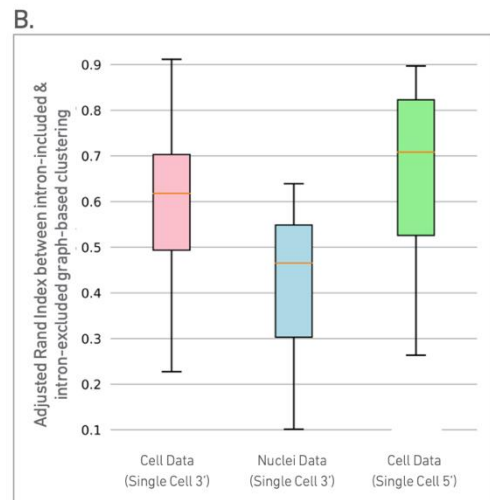
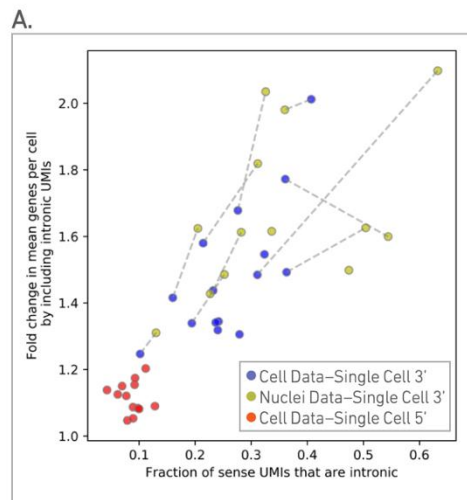
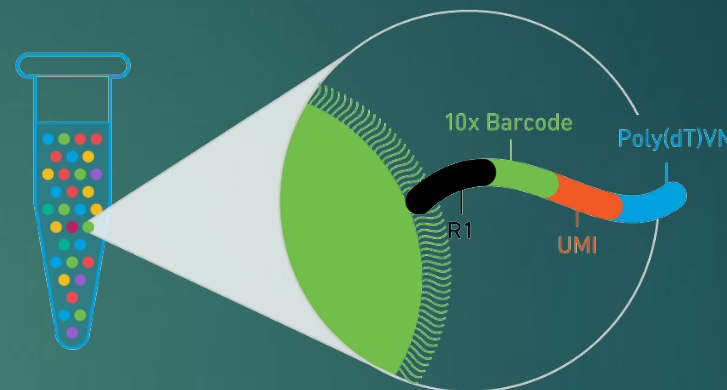


Assay Options: 5' vs. 3' Chemistry

5' bead

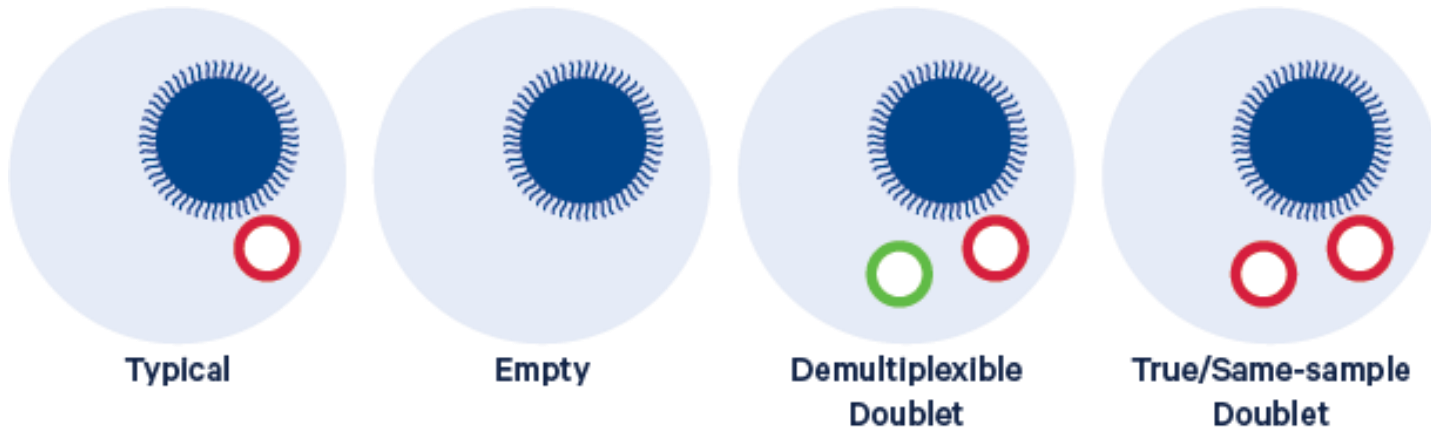
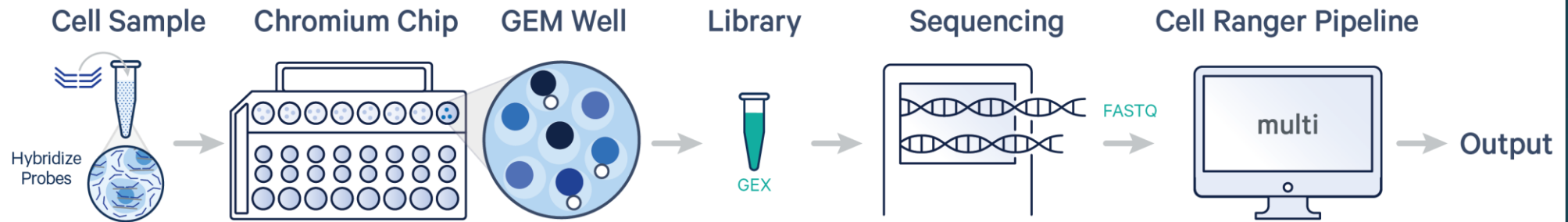


3' bead



And more! https://teichlab.github.io/scg_lib_structs/

Flex Chemistry



How deeply do you need to sequence?

Recommendations (reads/cell):

5'/3' GEM-X: 20K

(more with variant discovery)

Fixed RNA: 10-20K

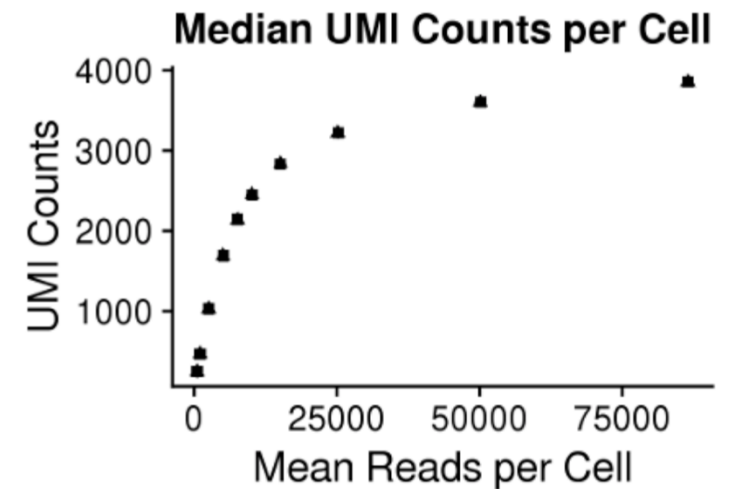
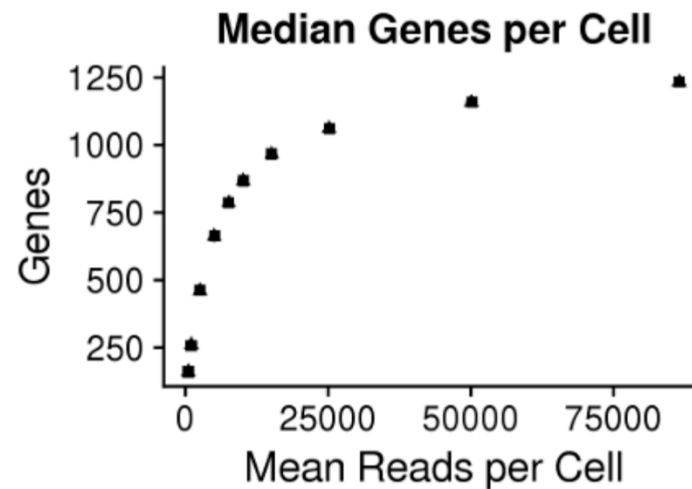
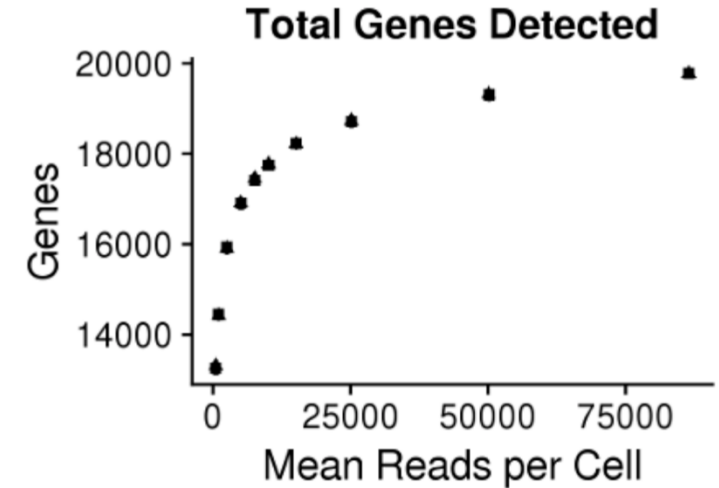
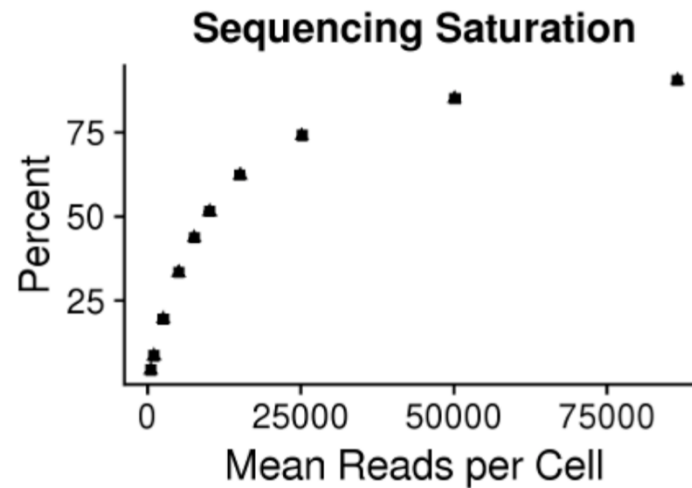
Different Recs for non GEX libraries

5' V(D)J - 5K

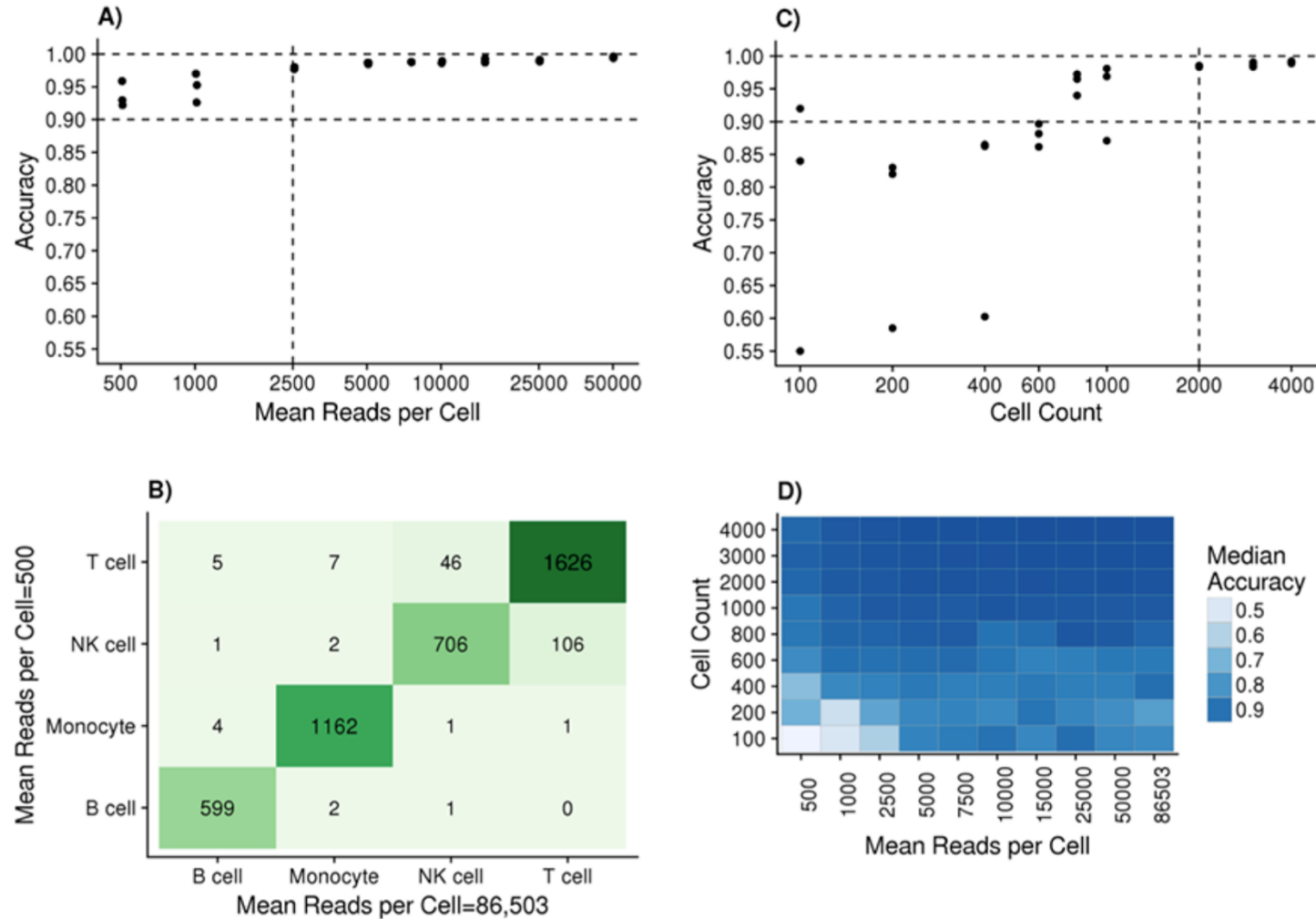
5' Cell Surface Protein- 5-20K

Hashtag only- 500

General Rule:
Achieve 90% saturation



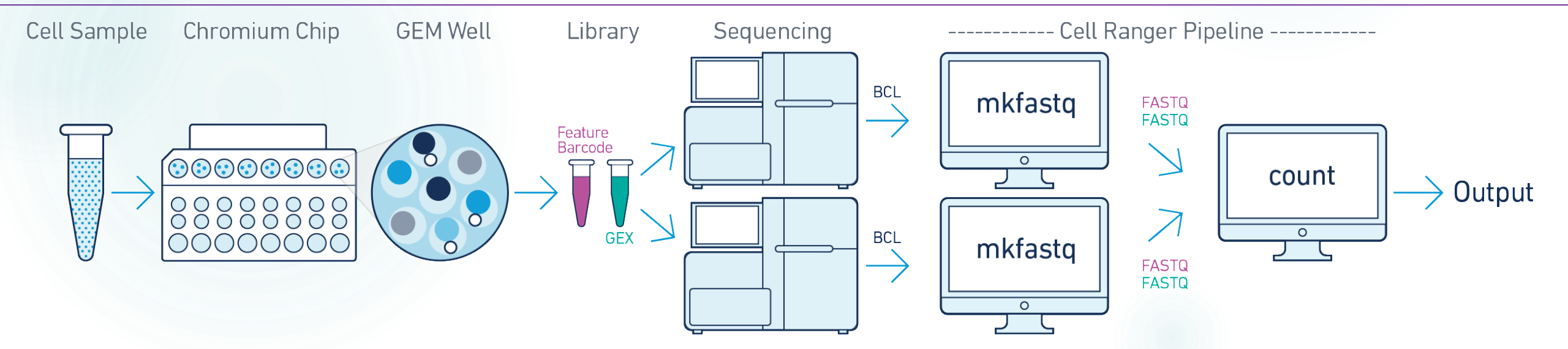
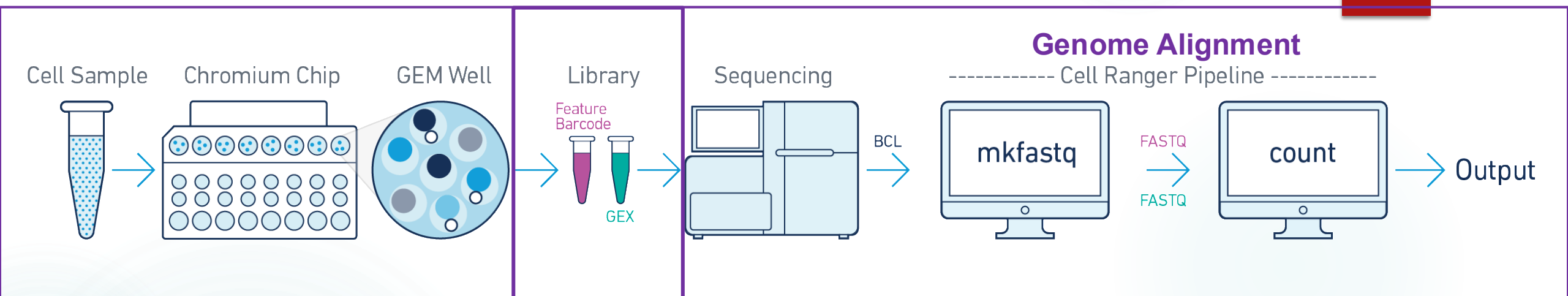
How deeply do you need to sequence?



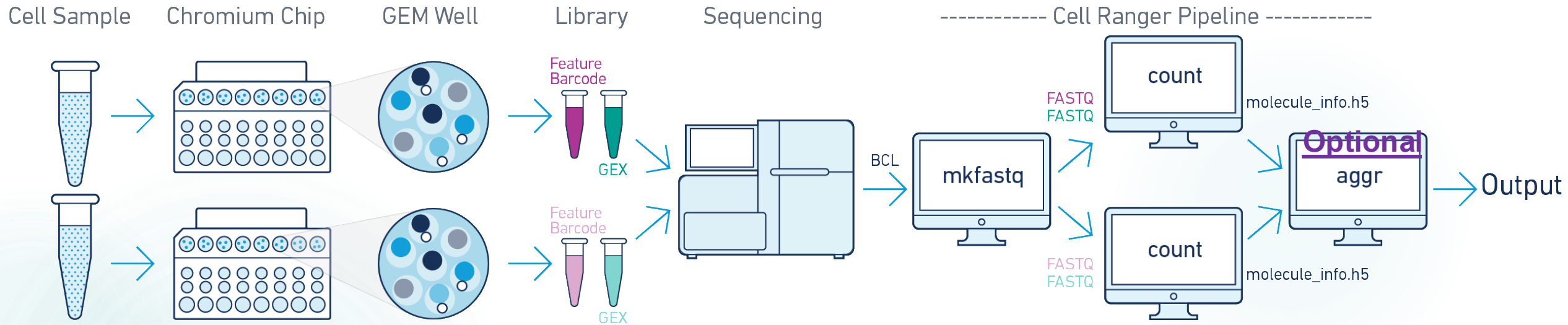
Sample Preparation: scRNA-seq

- ▶ To sort or not to sort, that is the question?
 - ▶ Factors to consider:
 - ▶ Do you have a population of interest in your sample or is this purely exploratory?
 - ▶ The number of cells present in your population of interest- too low prevalence is impossible to find
 - ▶ Keep in mind the doublet rate for all sample submissions: ~0.4% per 1000 cells
 - ▶ Will sorting interfere with downstream antibody staining?
 - ▶ What is the expected viability?
 - ▶ <https://satijalab.org/howmanycells/>

Sample Processing: MGI and Beyond



Sample Processing: Multiple Samples



- Two Options:
 - Aggregate samples in cellRanger
 - Randomly downsamples reads across samples to be equivalent (default setting-can be turned off)
 - Allows you to visualize multiple samples in the cloupe interface
 - Will need to have run Cellranger on individual samples first with exact same feature set
 - Barcodes used to identify cells will be appended with a number (e.g. -2, -3) that corresponds to order of aggregation
 - Merge samples/aggregate in R (or other packages)
 - Maintains original sequencing depth and tries to normalize sequencing by cells

Cell Ranger Important Info

- ▶ Genome Selection:
 - ▶ Human/mouse
 - ▶ Special Features- e.g. CARs
- ▶ Alignment for gene, protein (antibodies/hashtags), TCR/BCR, & ATAC fragments
- ▶ Output Key Files:
 - ▶ `Filtered_feature_bc_matrix`: input into most single-cell packages
 - ▶ In contrast to `raw_feature_bc_matrices`: this has all cell barcodes not just ones identified to be present in your data
 - ▶ Essentially a feature by cell matrix
 - ▶ `Web_summary.html`: user friendly visualization of QC metrics for cellranger
 - ▶ `.cloupe` file: point and click visual interface for limited single-cell analysis and visualization of data:
 - ▶ Outputs a umap/tsne of data, allows for visualization of genes and proteins, & subclustering of data, along with differential expression & graphs
 - ▶ Great way to explore the data and identify any potential problems with output quickly

Cell Ranger Web_summary

Summary Analysis

10,950

Estimated Number of Cells

67,137

Mean Reads per Cell

Sequencing

Number of Reads 735,154,190

Number of Short Reads Skipped 0

Valid Barcodes 86.1%

Valid UMIs 99.4%

Sequencing Saturation 88.2%

Q30 Bases in Barcode 93.4%

Q30 Bases in RNA Read 88.8%

Q30 Bases in RNA Read 2 85.6%

Q30 Bases in UMI 92.7%

Mapping

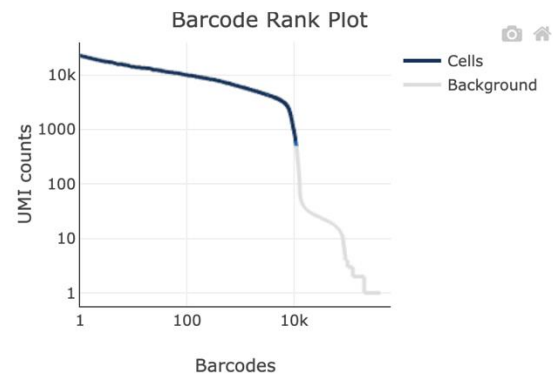
Reads Mapped to Genome 87.9%

Reads Mapped to Genome (GRCh38) 87.1%

Reads Mapped to Genome (mm10) 0.8%

Reads Mapped Confidently to Genome 71.5%

Cells



Estimated Number of Cells 10,950

Estimated Number of Cells (GRCh38) 10,856

Estimated Number of Cells (mm10) 1,784

Fraction Reads in Cells 95.5%

Fraction Reads in Cells (GRCh38) 95.6%

Fraction Reads in Cells (mm10) 40.6%

Mean Reads per Cell 67,137

Median Genes per Cell (GRCh38) 1,590

Median Genes per Cell (mm10) 2

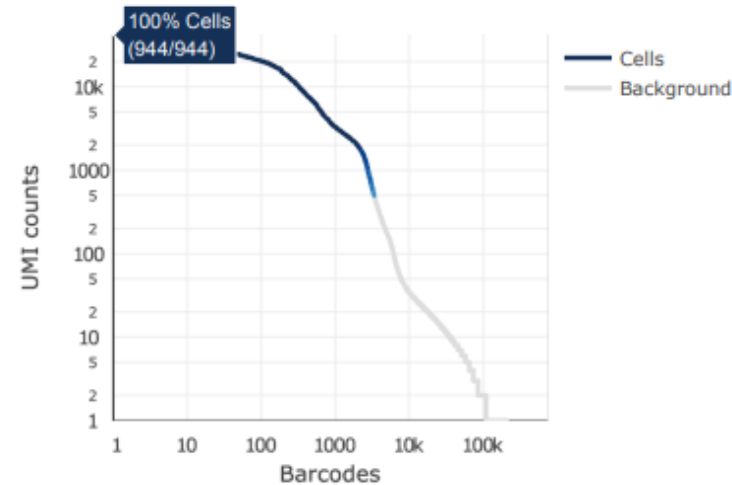
Total Genes Detected (GRCh38) 19,641

Total Genes Detected (mm10) 62

Median UMI Counts per Cell (GRCh38) 3,438

Median UMI Counts per Cell (mm10) 162

Cells



Sample

Sample ID Donor

Sample Description

Chemistry Single Cell 5' PE

Include introns False

Reference Path ...ata-cellranger-GRCh38-and-mm10-3.1.0

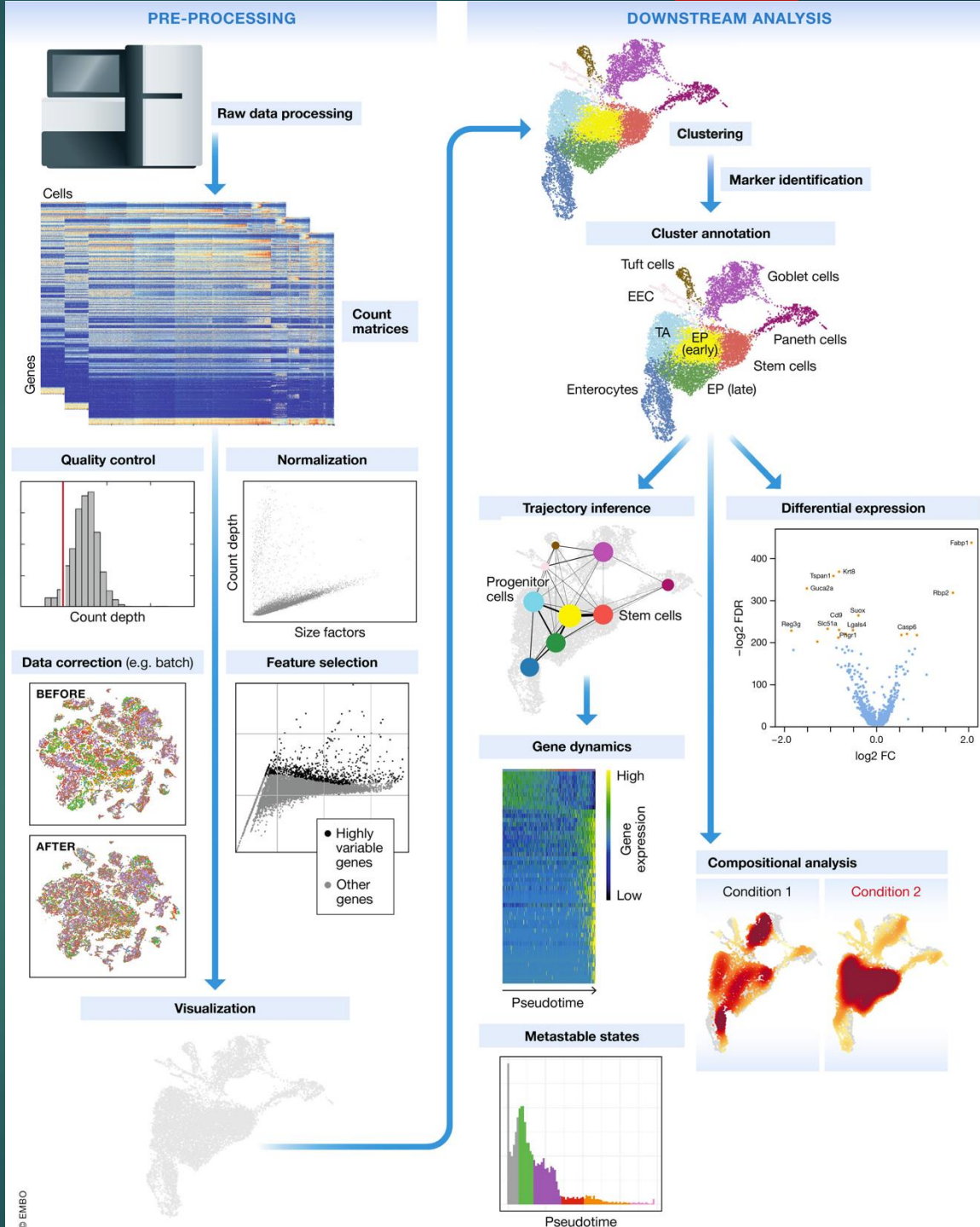
Transcriptome GRCh38_and_mm10-3.1.0

Pipeline Version cellranger-6.0.0

Possible reasons for low quality

Metric	Reason
Estimated Cells	Low viability, lysed cells
Target Reads/Cell	Rarely problematic
% Transcriptome mapping	Wrong transcriptome, low sequence quality
% Antisense Reads	Wrong chemistry, low sequence quality
Fraction reads in cells	Lysed cells, extracellular RNA

Sample Processing Post-Sequencing



Methods Galore

Number of single cell tools ~ Number of single cell studies ~ 500 (<https://www.scrna-tools.org/>)

Point and Click:

Loupe Browse (free)
Partek Flow (\$\$\$)
SeqGeq (\$\$\$)
kana (free)

General-purpose:

[Seurat V5](#)

[Scanpy](#)

Monocle V3
Scran
scater
LIGER (NMF)
scAlign
Scanorama
Kallisto bustools (replaces
cellranger)

Predicting the future:

RNAvelocity
Dynamo
CellRank
scVelo
Monocle 3 (R)
Slingshot (R)
PAGA (Python)
pCreode (Python)
STREAM
Palantir
CytoTrace (R/Python)

Large data sets:

scSVA
SAUCIE

Cell type assignment:

SingleR
CellHarmony
scPred
Moana
Garnet
Capybara

Mutation Detection:

CONICSmatrix (CNV)
HoneyBadger (CNV, LOH)
cb_sniffer (SNVs, Indels)
Vartrix (SNVs, Indels)

Analysis governed by two main principles

1. Single-cell RNA-seq data is very high-dimensional

2. And very sparse:

Fraction of transcripts captured per cell:

10x 3' V2: 14-15%

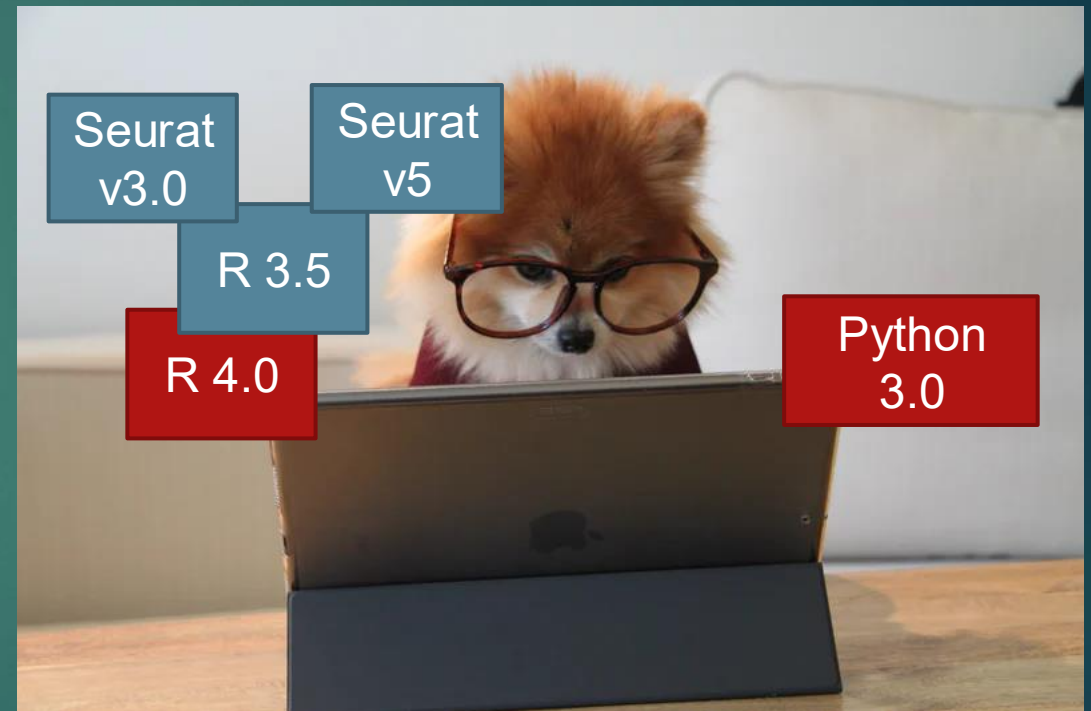
10x 3' V3: 30-32%

Additional Quirks:

- Transcripts encoding ribosomal proteins can comprise 30-50% of reads
- Top 100 transcripts often comprise ~50% of reads
- Low sensitivity
- Works best for cell type classification, more subtle signatures may get lost
- Results may favor highly expressed genes (e.g. *VIM*)
- *** Flex overcomes some of these issues

Seurat Analysis Package

- ▶ Most popular R language based package for analyzing scRNAseq, CITE-seq, hashtag, CRISPR-single cell data
- ▶ Essentially provides convenient wrappers around available methods for normalization, dimensional reduction, visualization, differential expression
- ▶ Available for windows & mac, however, I recommend dockers:



- Ability to run multiple versions, packages on the same system without uninstalling/re-installing.
- Can share the same "computer system" with colleagues to improve reproducibility

Practice Dataset

<https://www.10xgenomics.com/resources/datasets/integrated-gex-totalseq-c-and-bcr-analysis-of-chromium-connect-generated-library-from-10k-human-pbmcs-2-standard>

Inputs/Library

Single Channel

Peripheral blood mononuclear cells (PBMCs) were isolated from a healthy donor. Gene Expression, BCR, and Antibody Capture libraries were generated from a single channel of Chromium Connect. Antibody libraries were generated using BioLegend TotalSeq™-C TBNK panel. Libraries were prepared following the Chromium Next GEM Automated Single Cell 5' Reagent Kits v2 User Guide (CG0000507).

- 10,000 cells targeted
- 11,075 cells detected
- 68,396 mean reads per cell
- Sequenced on Illumina NovaSeq with 28 bp read 1, 90 bp read 2, 10 bp i5 sample barcode, and 10 bp i7 sample barcode
- Run with Cell Ranger 6.1.2

Published on April 20th, 2022

This dataset is licensed under the Creative Commons Attribution license.

Results Summary

View summary metrics about the sequencing quality and detected cells

View Summary

Download in browser

Batch download

If the file size is large, we suggest using [batch download](#) instead.

Output Files	Size	md5sum
Summary HTML	4.92 MB	658b03d02081e4749f93f509637c5b7a
Summary CSV	6.1 kB	529933a6b4ae61effa602ef0fc211e79
Gene Expression - Loupe Browser file	114 MB	7b3fd33a921d80b13f7d76221a562142
Gene Expression - Genome-aligned BAM	41.7 GB	29d7a7d08af23b9150c8d0792acd6ddf
Gene Expression - Genome-aligned BAM index	10.2 MB	5b4509d1afd3d503b3b63a5db93e9464
Gene Expression - Sample barcodes	288 kB	05db5e31ca509d3549d54f829d54a8d9
Gene Expression - Feature / cell matrix HDF5 (per-sample)	31.9 MB	192eb0a882b8ebe89830d76bcecad45c
Gene Expression - Feature / cell matrix (per-sample)	83.9 MB	1d63cb8de80eea2e957fa6da55c7dd7c



Seurat-v5 Basics

1. In R terminal, type:

?FunctionName

example: ?FindAllMarkers

2. Code available on github, e.g.:

<https://github.com/satijalab/seurat/blob/master/man/FindClusters.Rd>

```
object.data <- Read10X("outs/filtered_feature_bc_matrix/")
```

- ▶ File path must have EXACTLY filtered_feature_bc_matrix Or with newer versions, can be sample_feature_bc_matrix
- ▶ This reads in a large matrix into R- if you have multiple data modalities, there will be a matrix for each modality (e.g. Gene, antibodies)
- ▶ These can be accessed via the \$ e.g. object\$`Gene Expression` - Note that R needs the `` to handle the spaces in your assay

```
object <- CreateSeuratObject(object.data, project = "your project name")
```

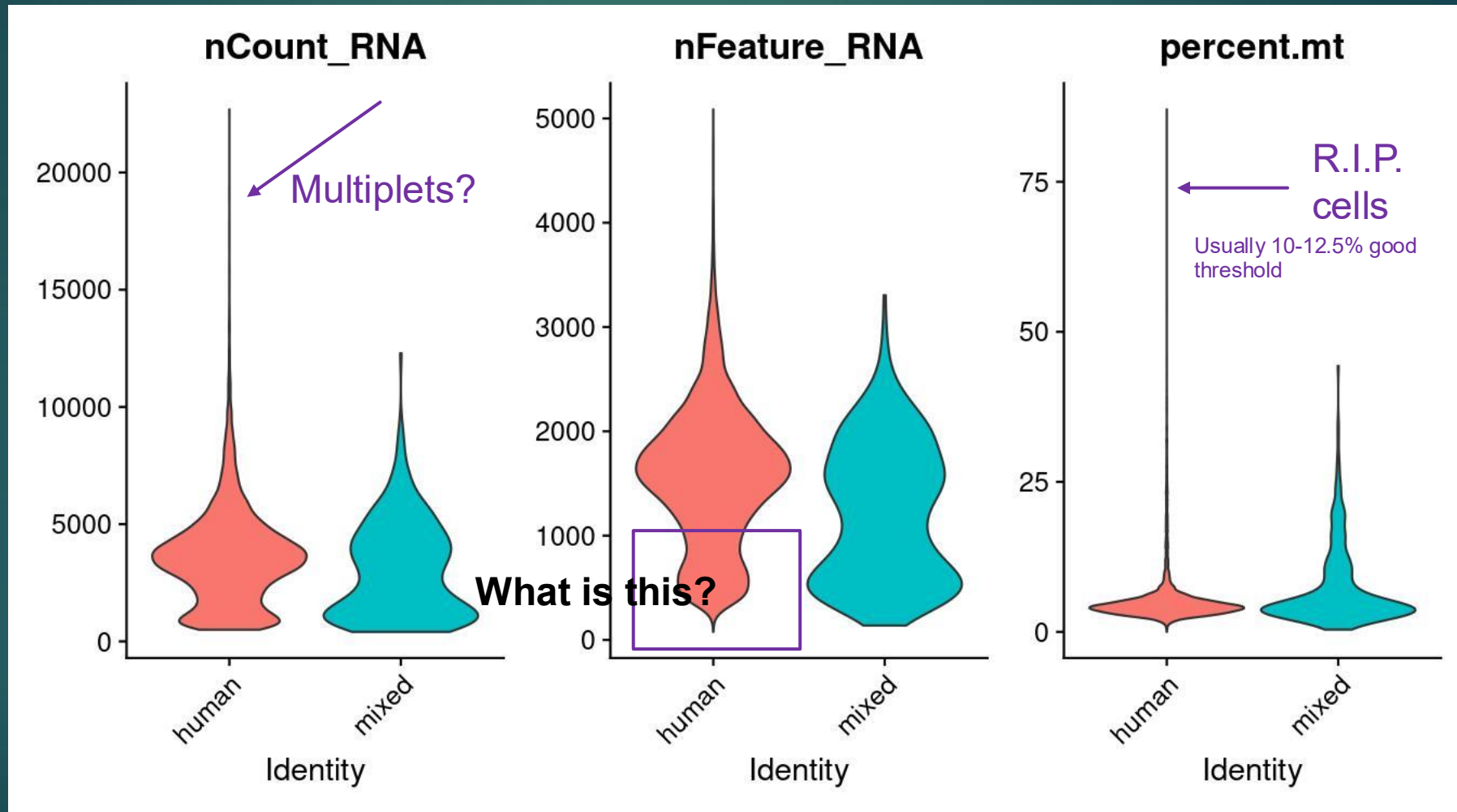
- ▶ If you have multiple data modalities you will need to replace object.data with object.data\$`Gene Expression`
- ▶ You can remove features here- **how does this align with your research question & analysis and downstream algorithms?**

```
object[["percent.mt"]] = PercentageFeatureSet(object, pattern = "^MT-", assay = "RNA")
```

- ▶ In human genome, all mitochondrial genes are prefixed with "MT", and exclusion of dead cells is an important step in QC so we want to understand what percentage of counts in each cell are due to mitochondrial genes
- ▶ This tells R to look for genes that **start** with MT- note this is for human and mouse genes are named differently
- ▶ Some people will also look at ribosomal features

Seurat-v5

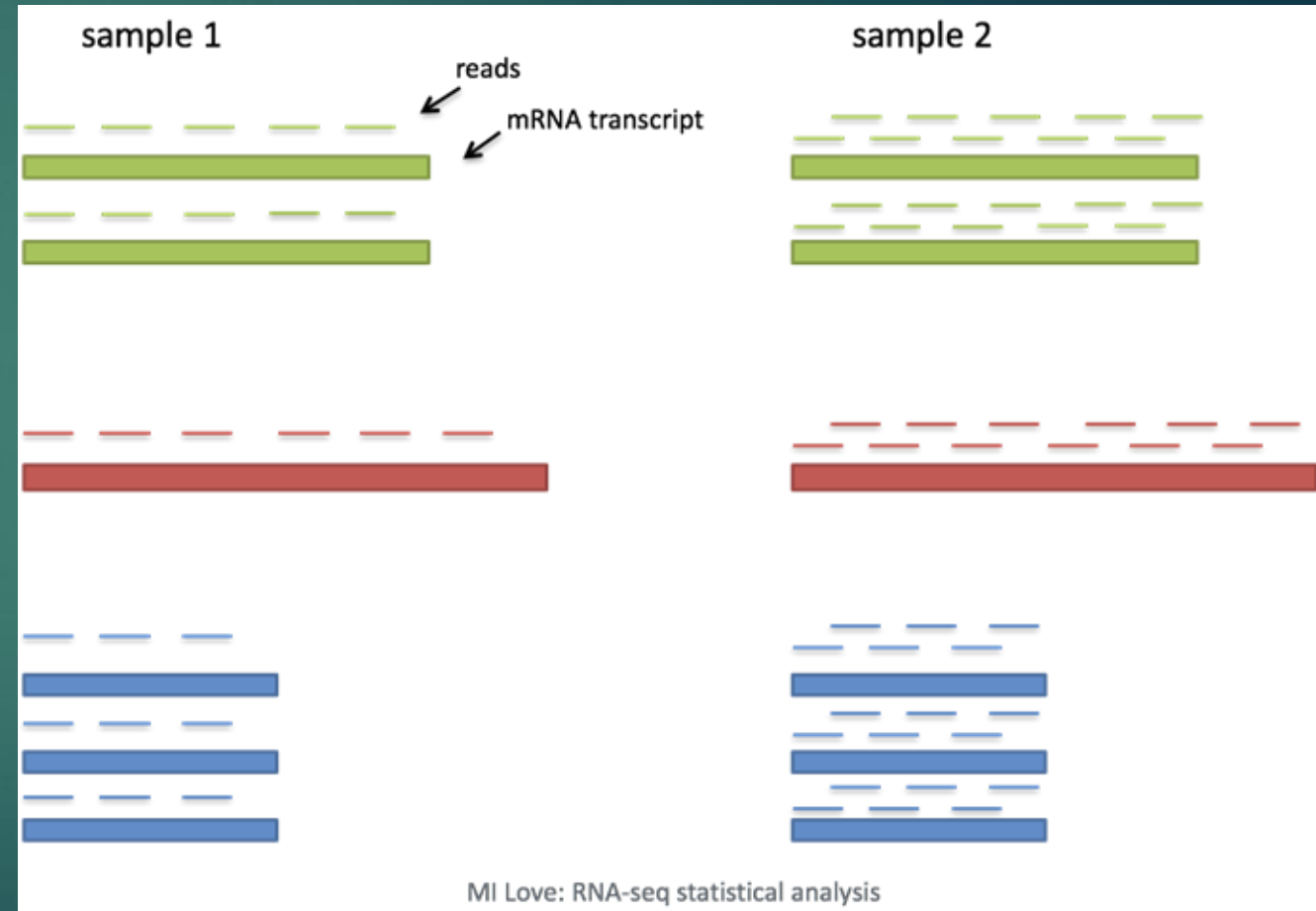
```
VlnPlot(object, features = c("nCount_RNA", "nFeature_RNA", "percent.mt"), ncol = 3, pt.size = 0)
```



```
object <- subset(object, nFeature_RNA > 200 & percent.mt < 10 & nFeature_RNA < 5000 & nCount_RNA < 15000)
```

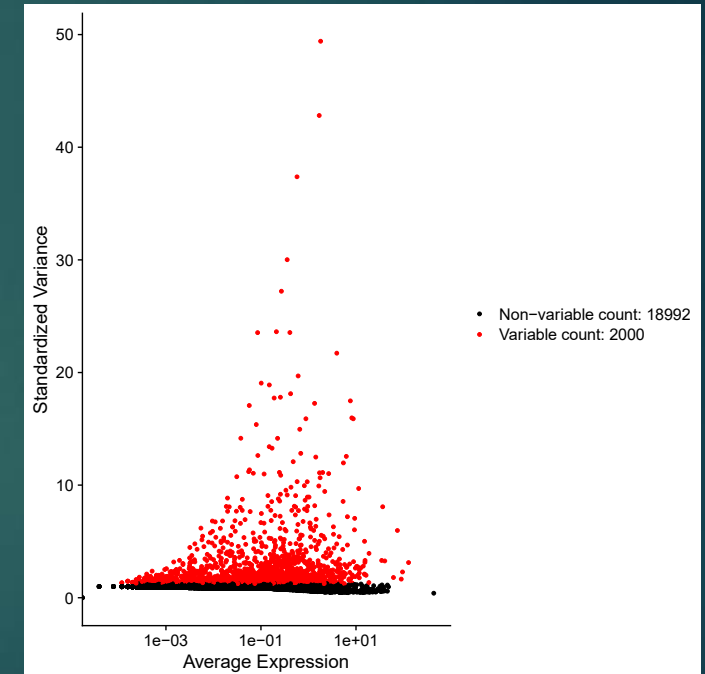
Normalize, scale, control for unwanted variation

- Goal: remove technical effects while preserving biological variation
- Sequencing depth of a cell: total UMI (nCounts)
- Even in the same experiment, different cells have very different sequencing depths
- Expression level of a gene in a cell is proportional to the sequencing depth of the cell, unless the data is normalized to sequencing depth
- Normalize all cells to equal levels; HOWEVER, this assumes all cells are the same size
 - **NormalizeData:**
 - Divide by total counts in each cell
 - Scale to fixed counts (default is 1×10^4 use 1×10^6 for CPM)
 - Add 1
 - natural log

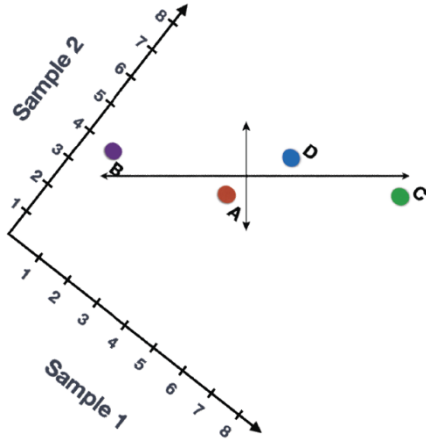


FindVariableFeatures

- **FindVariableFeatures:** use a variance stabilizing transformation
- **vst:** Variance-Stabilizing Transformation
 - For each gene, plot $\log(\text{Var})$ vs $\log(\text{mean})$
 - Use loess linear regression to calculate expected variance based on observed mean
 - Standardize the expression of each gene so that its variance matches the calculated expected variance
 - Plot standardized variance vs $\log(\text{mean})$ and choose outliers
 - Choose 2000-3000 outliers
- `mean.var.plot`
- Dispersion
- **ScaleData:** subtract mean, divide by standard deviation, remove unwanted signal using multiple regression
 - Not always: Should genes be weighted equally or is the magnitude of expression convey importance of a gene?

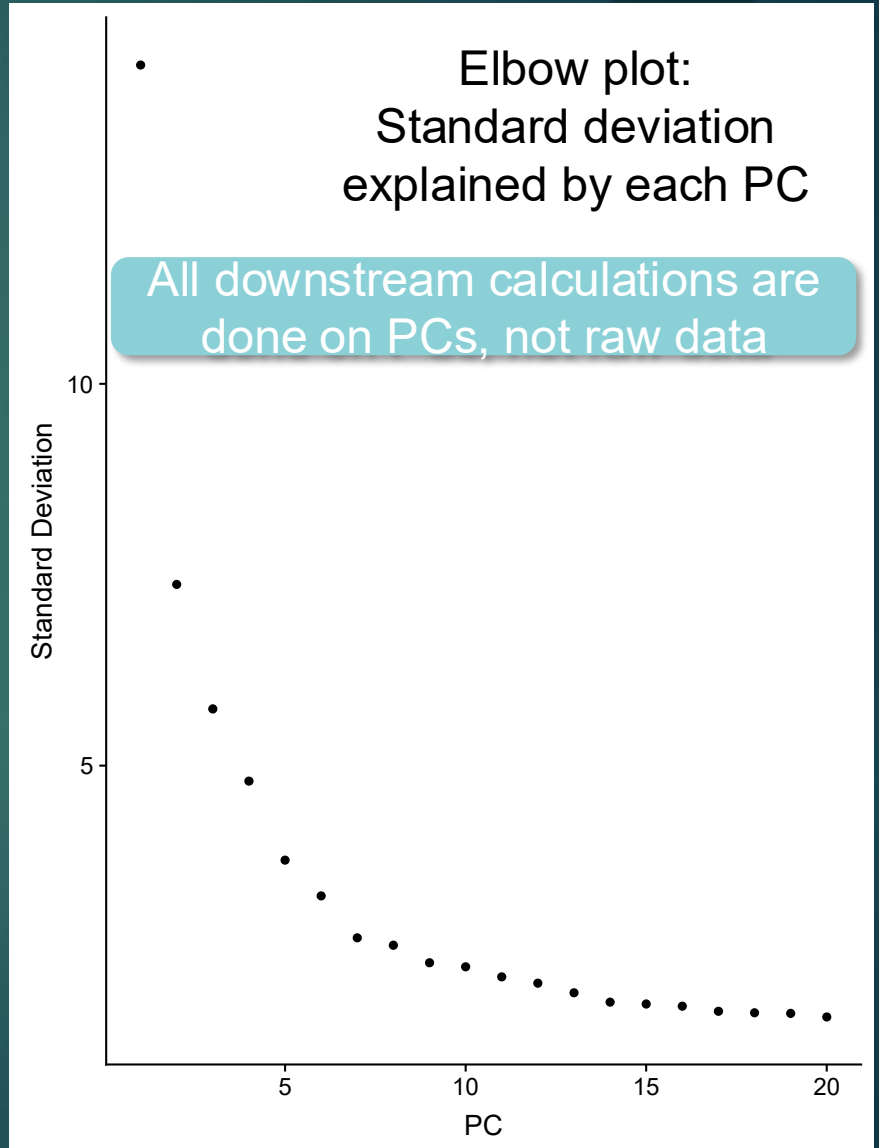


Selecting Principal Components

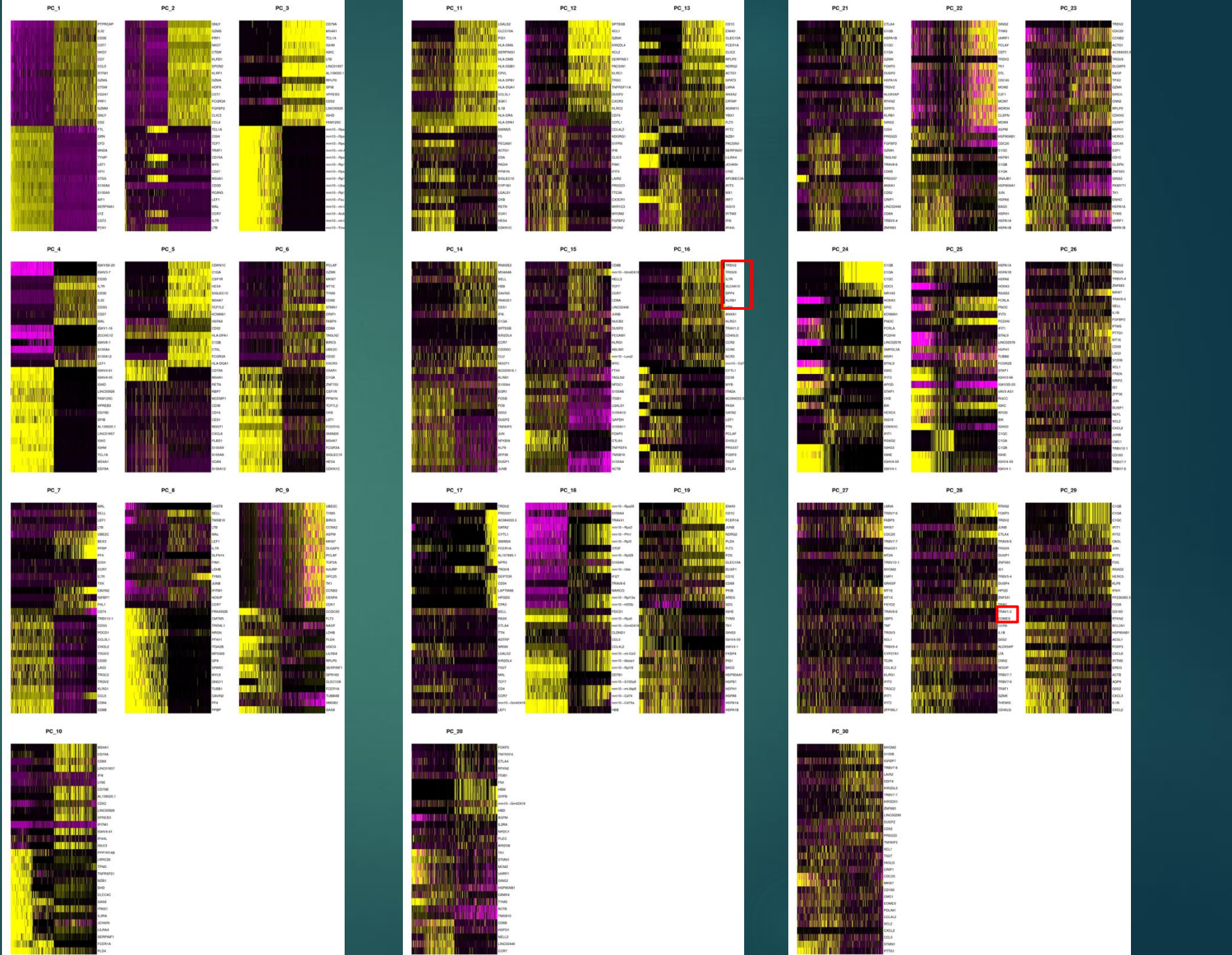


	Sample 1	Sample 2	Influence on PC1	Influence on PC2
Gene A	4	5	-2	0.5
Gene B	1	4	-10	1
Gene C	8	8	8	-5
Gene D	5	7	1	6

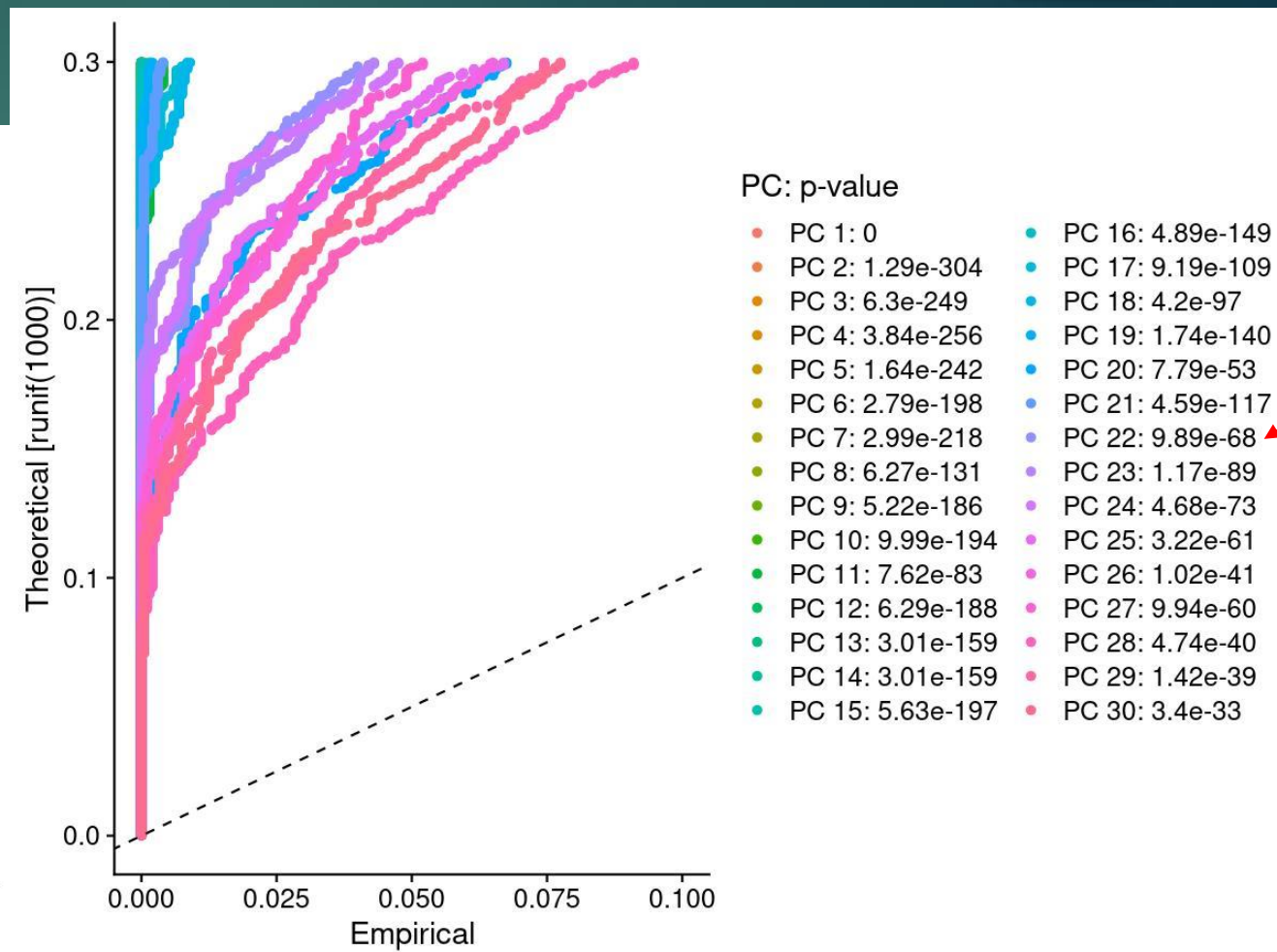
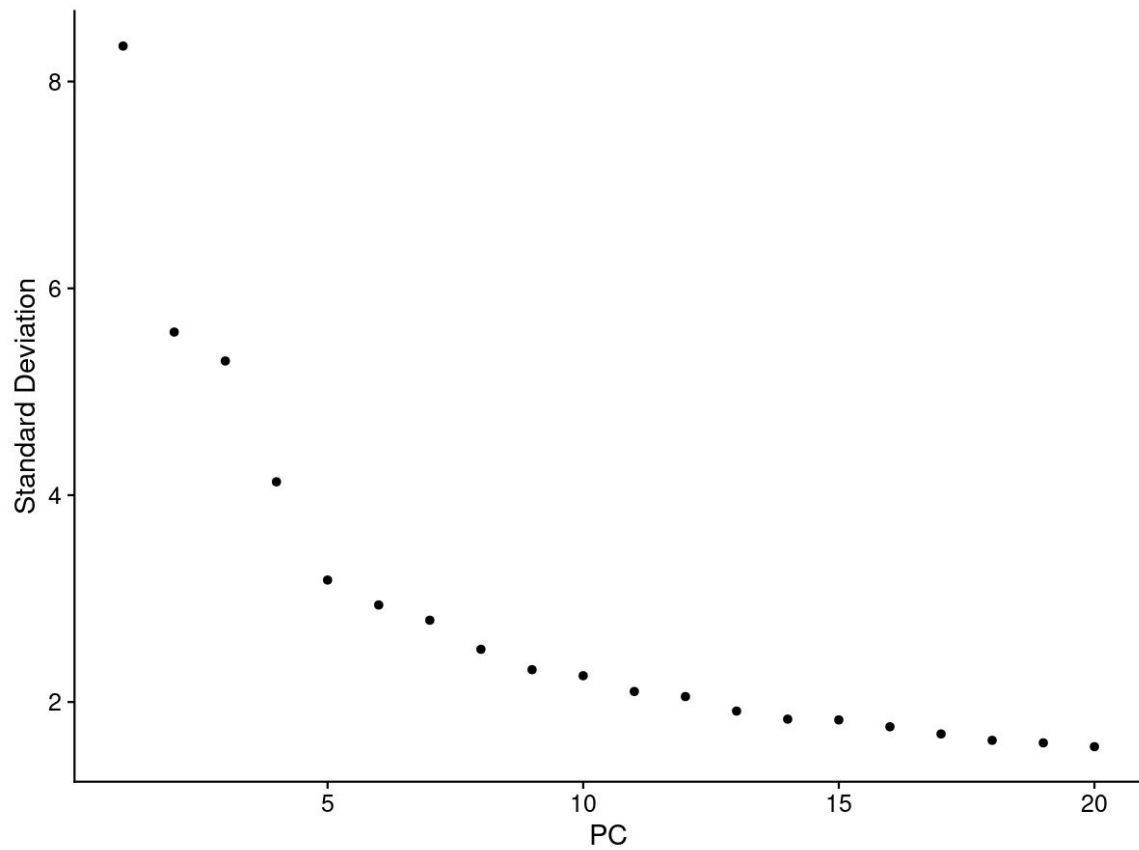
- Retain and plot key information about each principal component (PC):
 - Percentage of standard deviation explained
 - P-value (obtained from bootstrapping “Jackstraw”)
 - Plot gene expression heatmaps for each of the top ~12 principal components
- Choose Principal Components:
 - Purpose: choose relative importance of minor expression signatures
 - Discontinuity in elbow plot
 - All PCs that explain $\geq x\%$ of SD (e.g. 2%)
 - P-value from JackStraw analysis $< 1 \times 10^{-n}$ (e.g. 1×10^{-100})
 - Clarity of PC heatmaps

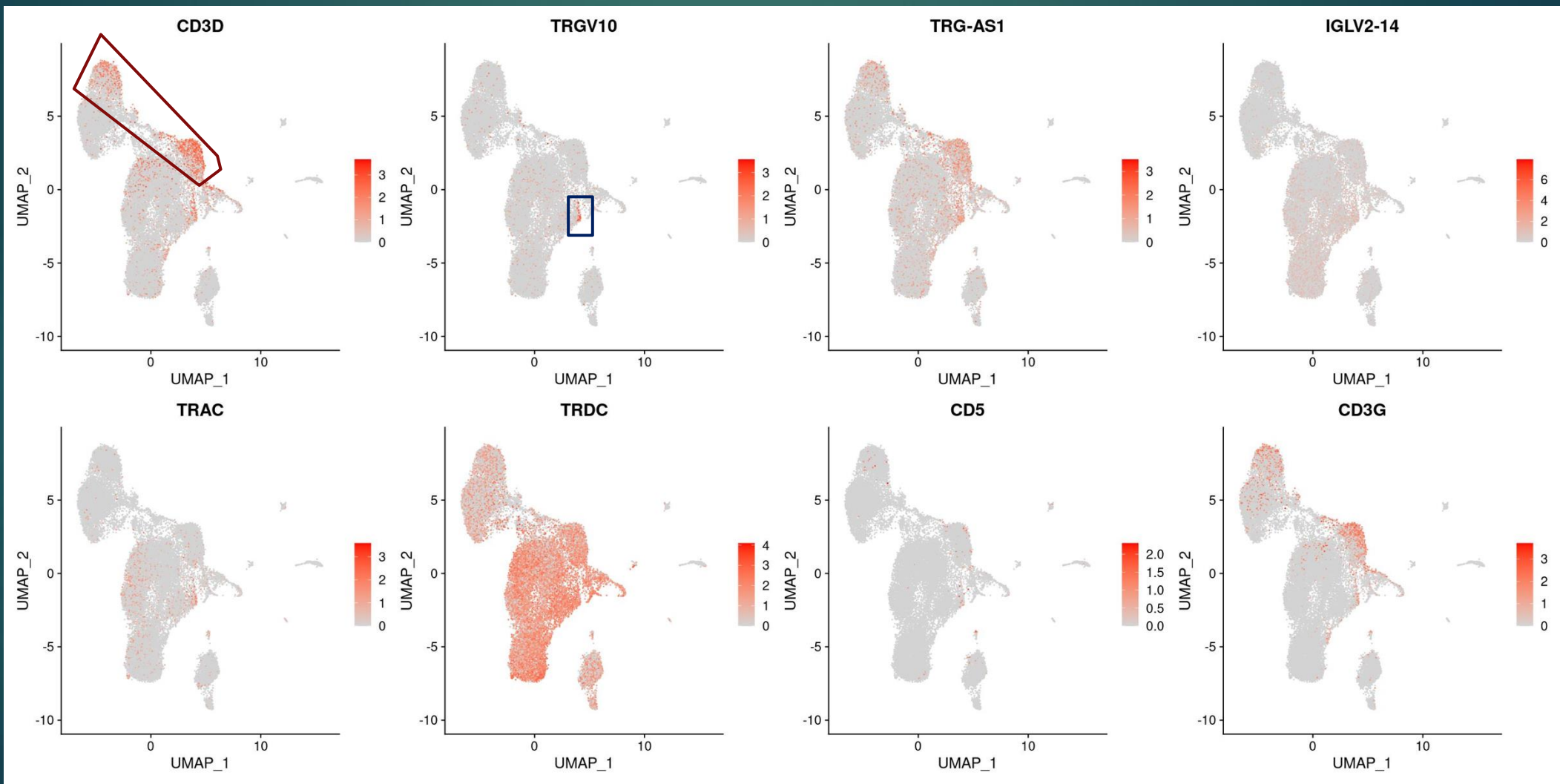


Dim Heatmaps

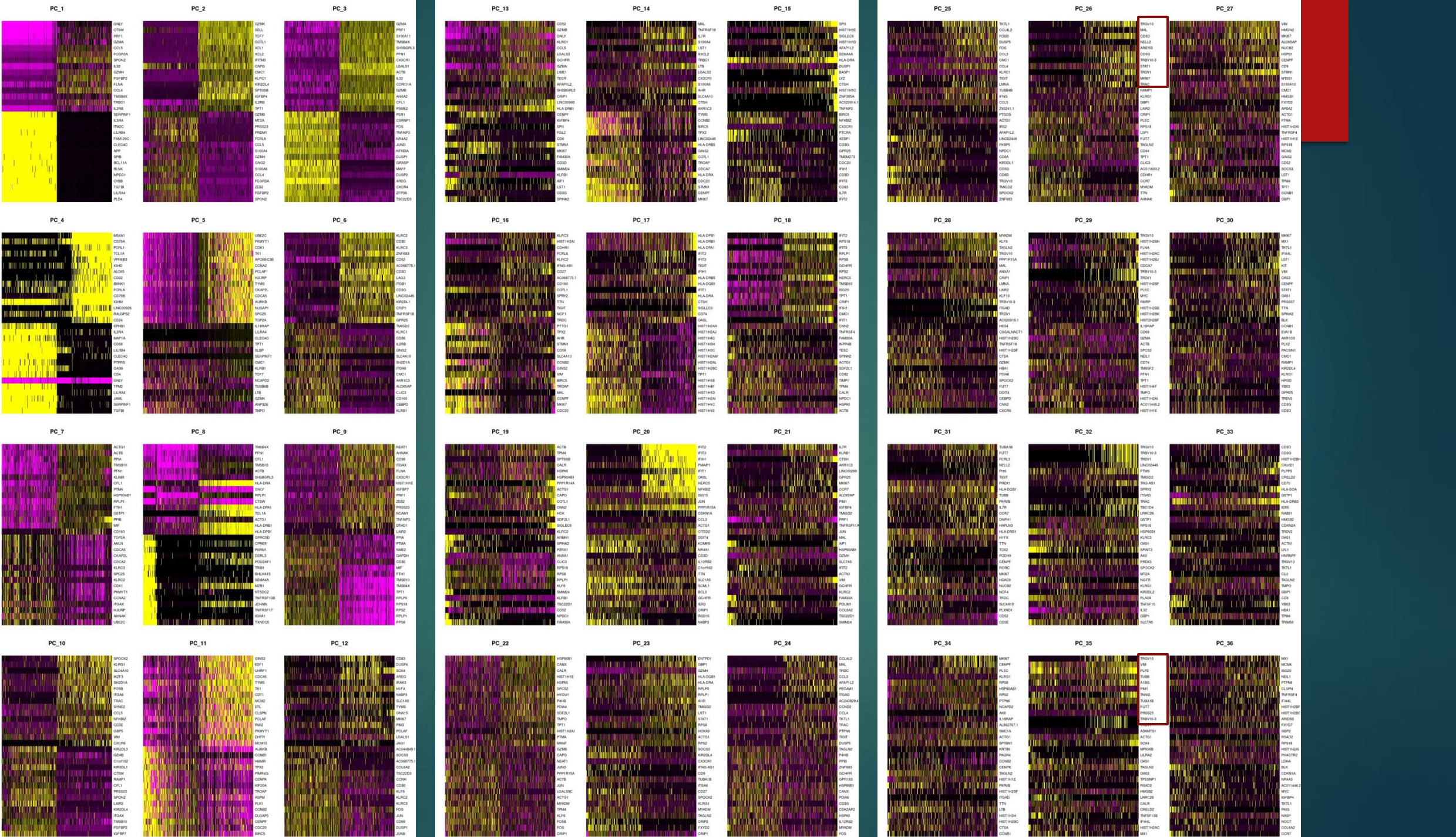


PC Demo

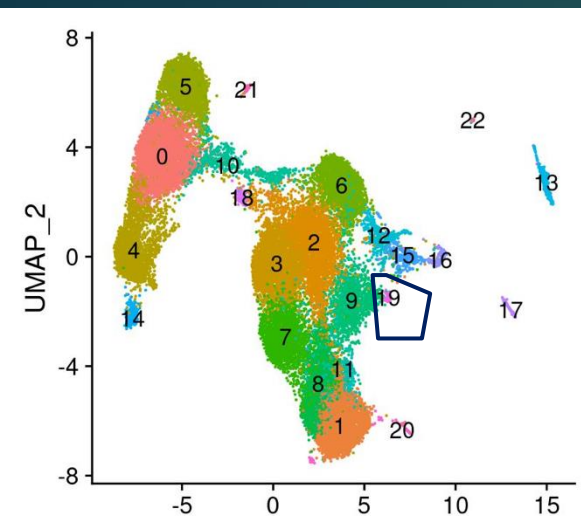




With various cluster resolutions (up to 2.5), the best I could do was get 25% of the cluster to be TRGV10+



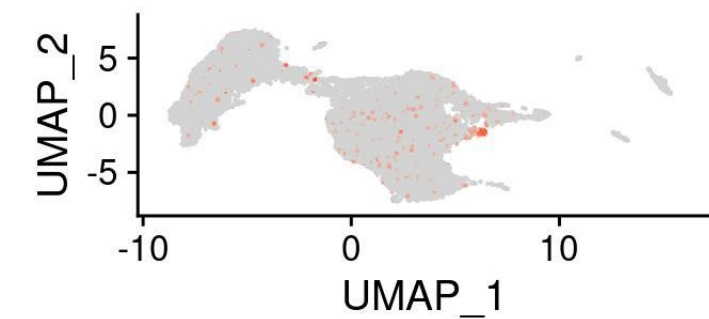
1.0 res



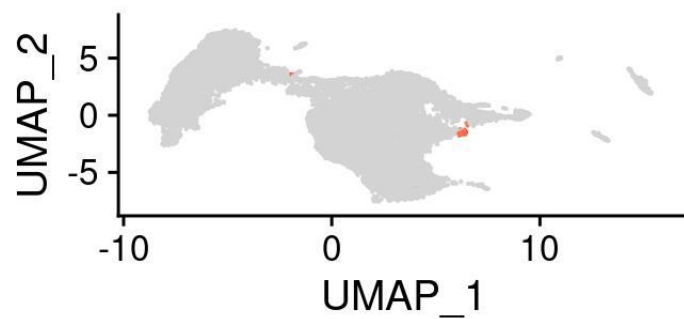
PC35



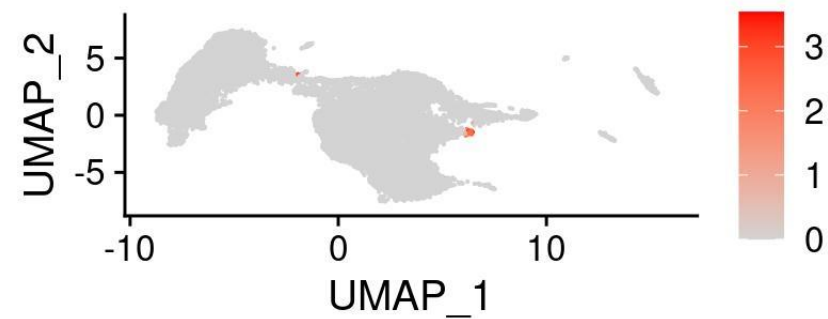
TRGV10



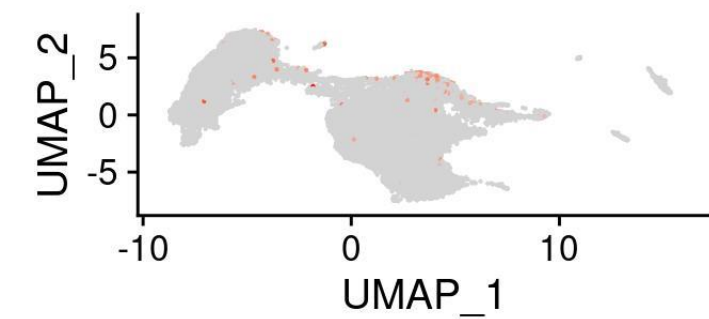
TRBV10-3



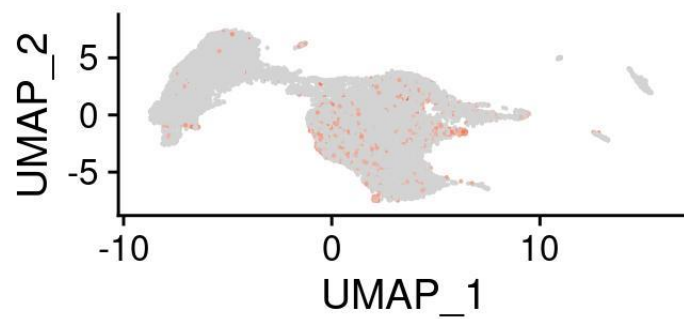
TRDV1



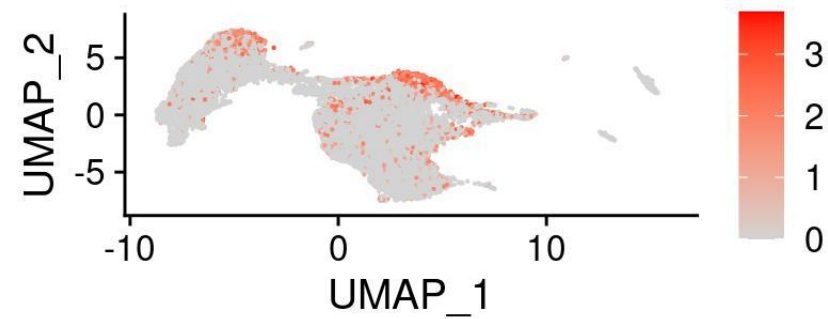
TRDV2



TRAC

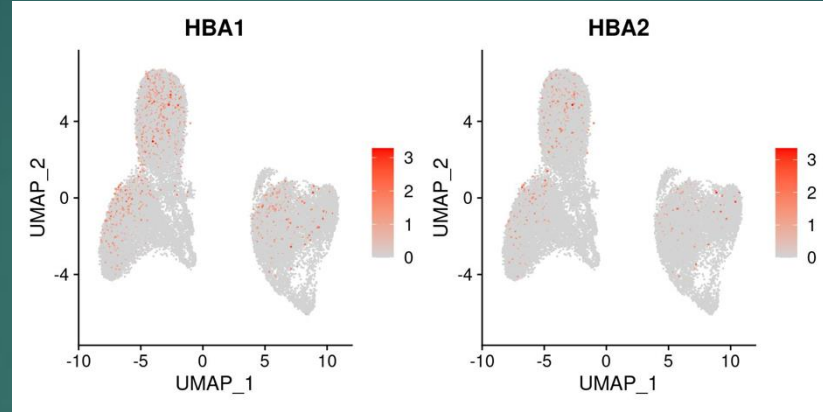


CD3D

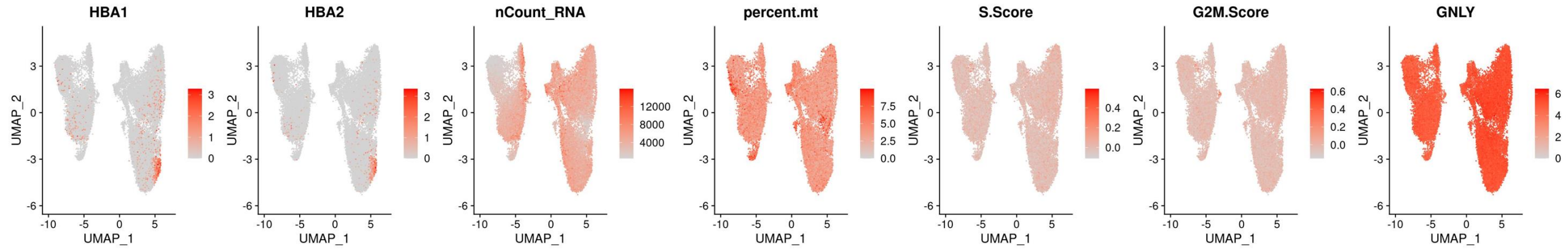


Is more always better?

NK Cells PC14

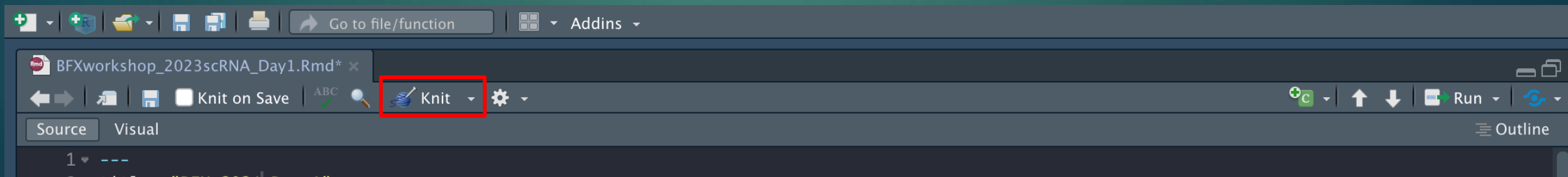


PC50



Homework

- ▶ Run Rmarkdown document for this week
 - ▶ 1st download the dataset specified earlier
 - ▶ Change the directory path in the R markdown document to where the file is located on your computer
 - ▶ Read through the script, noting you only need to change the top variables if you'd like to (recommended to gain experience in how they can affect your data)
 - ▶ Then Knit to create a html print out of your work!



Next week: Differential expression, batch correction, & handling CITE-seq data

Resources

- ▶ <https://rnabio.org/module-08-scrna/0008/02/01/scRNA/>
- ▶ <http://bioconductor.org/books/release/OSCA/>
- ▶ https://hbctraining.github.io/scRNA-seq_online/lessons/01_intro_to_scRNA-seq.html
- ▶ PCA: <https://www.youtube.com/watch?v=FgakZw6K1QQ&t=0s>
- ▶ <https://www.sc-best-practices.org/preamble.html>